

RQ Answers

omitted

2024-04-05

Data overview

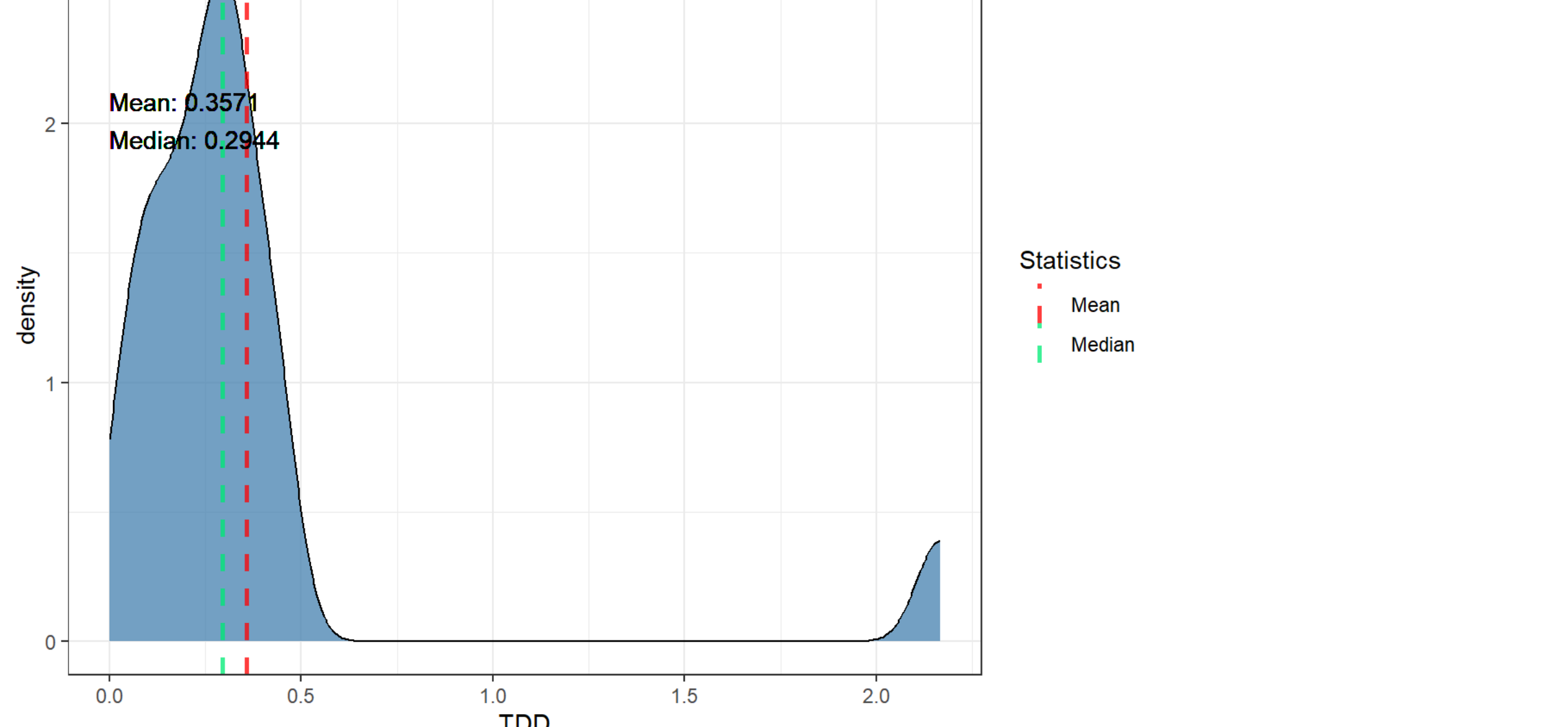
glimpse(all_data)				
## Rows: 639				
## Columns: 11				
## \$ repo	<chr> "commons-collections", "commons-collections", "com...			
## \$ pr_number	<int> 203, 203, 203, 203, 203, 203, 203, 203, 203, ...			
## \$ rule	<chr> "java:S3776", "java:S3358", "java:S3358", "java:S2...			
## \$ severity	<chr> "CRITICAL", "MAJOR", "MAJOR", "MAJOR", "MAJOR", "I...			
## \$ file	<chr> "src/main/java/org/apache/commons/collections4/com...			
## \$ type	<chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "BUG", "O...			
## \$ status	<chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O...			
## \$ debt	<int> 11, 5, 5, 15, 2, 0, 30, 10, 12, 10, 10, 10, 10...			
## \$ nloc_affected_file	<int> 108, 108, 108, 108, 108, 57, 194, 194, 194, 194, 1...			
## \$ complexity	<int> 37, 37, 37, 37, 8, 43, 43, 43, 43, 43, 43, 43,...			
## \$ origin	<chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P...			

RQ1

TDD (Technical Debt Density) by PR Distribution

## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the data has 17 rows.	
## i Please consider using `annotate()` or provide this layer with data containing	
## a single row.	

## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but the data has 17 rows.	
## i Please consider using `annotate()` or provide this layer with data containing	
## a single row.	

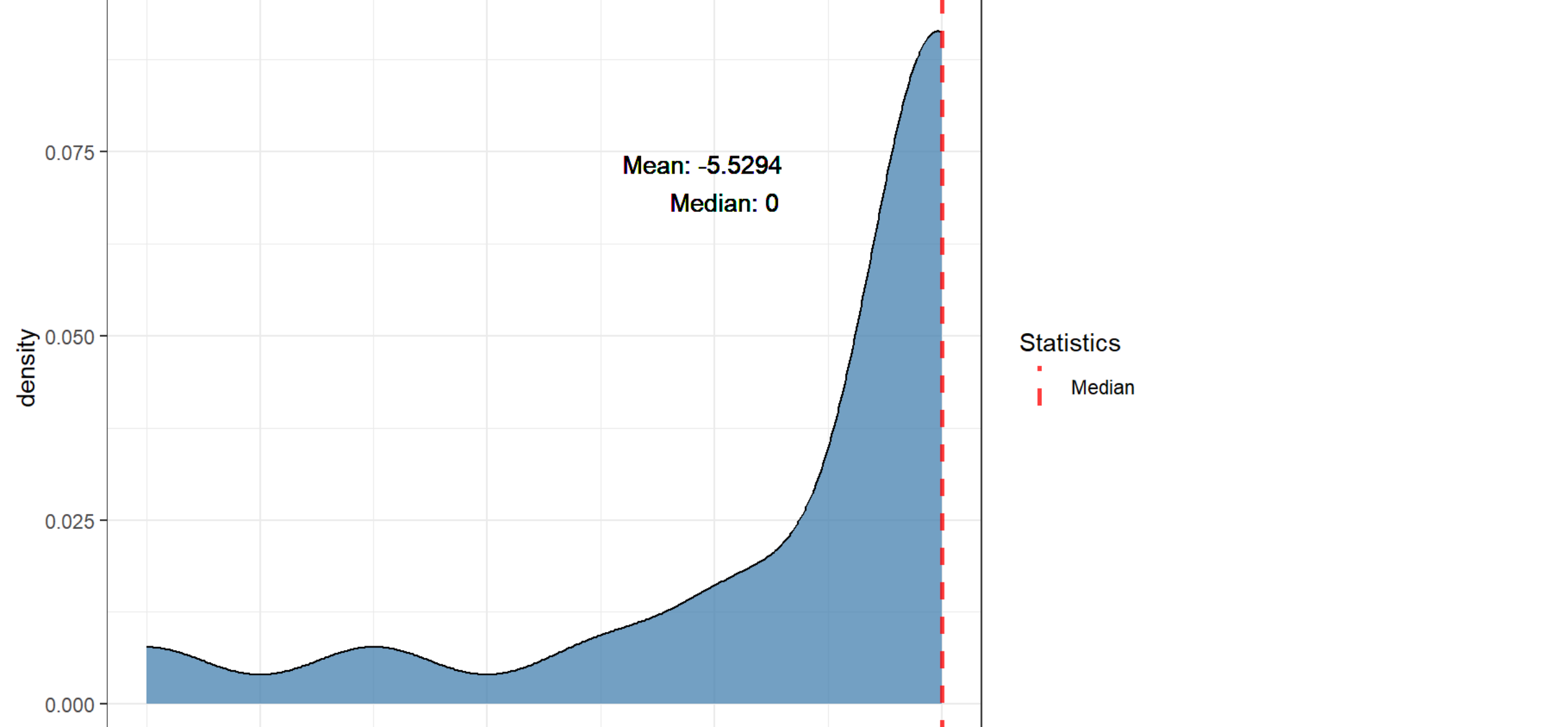


TDV (Technical Debt Variation)

TDV Distribution

## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the data has 17 rows.	
## i Please consider using `annotate()` or provide this layer with data containing	
## a single row.	

## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but the data has 17 rows.	
## i Please consider using `annotate()` or provide this layer with data containing	
## a single row.	



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1 commons-collectio...	35.3	64.7		0

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3				
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero		
## <dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	35.3	64.7		0

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3				
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_th...		
## <chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1 commons-collections	58.8		41.2	
## i abbreviated name: 'preexisting_tdv_greater_than_zero				

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2				
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero			
## <dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	58.8		41.2	

Percentages for new TDV by repo

## # A tibble: 1 × 3				
## Groups: repo [1]				
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero		
## <chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1 commons-collections	0		100	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2				
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero			
## <dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	0		100	

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_j$$

given a rule *i*, its *PositionScore_i* will be the sum of its position across all *k* repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 8 × 2				
## # Groups: rank [8]				
## rank 'commons-collections'				
## <int> <chr>				
## 1	1 java:S1125	-	8	
## 2	2 java:S1197	-	5	
## 3	3 java:S1612	-	2	
## 4	4 java:S1192	-	1	
## 5	5 java:S1155	-	1	
## 6	6 java:S1128	-	1	
## 7	7 java:S2583	-	1	
## 8	8 java:S1602	-	1	

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

## # A tibble: 8 × 2				
## rule	position_score			
## <chr>	<dbl>			
## 1 java:S1125	122			
## 2 java:S1197	123			
## 3 java:S1612	124			
## 4 java:S1128	125			
## 5 java:S1155	126			
## 6 java:S1192	127			
## 7 java:S1602	128			
## 8 java:S2583	129			

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2				
## # Groups: rank [10]				
## rank 'commons-collections'				
## <int> <chr>				
## 1	1 java:S1125	-	87	
## 2	2 java:S128	-	52	
## 3	3 java:S2160	-	52	
## 4	4 java:S2065	-	51	
## 5	5 java:S3776	-	46	
## 6	6 java:S1192	-	41	
## 7	7 java:S131	-	30	
## 8	8 java:S1452	-	20	
## 9	9 java:S1948	-	20	
## 10	10 java:S2259	-	19	

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

## # A tibble: 10 × 2				
## rule	position_score			
## <chr>	<dbl>			
## 1 java:S1125	122			
## 2 java:S128	123			
## 3 java:S2160	124			
## 4 java:S2065	125			
## 5 java:S3776	126			
## 6 java:S1192	127			
## 7 java:S131	128			
## 8 java:S1452	129			
## 9 java:S1948	130			
## 10 java:S2259	131			

Outliers

Higher TDD

## # A tibble: 17 × 5					
## repo	pr_number	total_debt	total_nloc	tdt	
## <chr>	<int>	<int>	<int>	<dbl>	<dbl>
## 1 commons-collections	259	340	157	2.17	
## 2 commons-collections	67	53	119	0.445	
## 3 commons-collections	260	5	12	0.417	
## 4 commons-collections	307	1188	3021	0.393	
## 5 commons-collections	65	320	987	0.324	
## 6 commons-collections	208	2643	8168	0.324	
## 7 commons-collections	203	956	3159	0.303	
## 8 commons-collections	274	89	301	0.296	
## 9 commons-collections	407	443	1505	0.294	
## 10 commons-collections	300	126	541	0.233	
## 11 commons-collections	362	231	999	0.231	
## 12 commons-collections	69	529	2863	0.185	
## 13 commons-collections	316	28	169	0.166	
## 14 commons-collections	317	17	152	0.112	
## 15 commons-collections	315	27	281	0.0961	
## 16 commons-collections	329	19	301	0.0631	
## 17 commons-collections	361	2	86	0.0233	

Higher TDV

## # A tibble: 17 × 3				
## pr_number	repo	tdv		
## <int>	<chr>	<int>		
## 1	65 commons-collections	0		
## 2	203 commons-collections	0		
## 3	208 commons-collections	0		
## 4	259 commons-collections	0		
## 5	260 commons-collections	0		
## 6	300 commons-collections	0		
## 7	307 commons-collections	0		
## 8	316 commons-collections	0		
## 9	317 commons-collections	0		
## 10	329 commons-collections	0		
## 11	407 commons-collections	0		
## 12	361 commons-collections	-2		
## 13	315 commons-collections	-8		
## 14	274 commons-collections	-9		
## 15	67 commons-collections	-15		
## 16	362 commons-collections	-25		
## 17	69 commons-collections	-35		

Lower TDV

## # A tibble: 17 × 3				
## pr_number	repo	tdv		
## <int>	<chr>	<int>		
## 1	69 commons-collections	-35		
## 2	362 commons-collections	-25		
## 3	67 commons-collections	-15		
## 4	274 commons-collections	-9		
## 5	315 commons-collections	-8		
## 6	361 commons-collections	-2		
## 7	65 commons-collections	0		
## 8	203 commons-collections	0		
## 9	208 commons-collections	0		
## 10	259 commons-collections	0		
## 11	260 commons-collections	0		
## 12	300 commons-collections	0		
## 13	307 commons-collections	0		
## 14	316 commons-collections	0		
## 15	317 commons-collections	0		
## 16	329 commons-collections	0		
## 17	407 commons-collections	0		

Extra

Number of PRs with pre-existing TD

## n	
## 1 17	

Number of java:S1192 issues that affect test code

## n	
## 1 0	

Number of java:S1192 issues that affect not test code

## n	
## 1 42	

Summary NCLOC

## nloc	
## Min. : 12	
## 1st Qu.: 157	
## Median : 301	
## Mean :1340	
## 3rd Qu.:1505	
## Max. :8128	